reaches the earth from outer space, and in the atmospheric envelope of the earth it releases every possible sort of secondary radiation, akin in character to radioactive radiation. However, it is only since 1947 that we have known anything at all about the cause of this remarkable phenomenon. This knowledge is the fruit of the work of Forbush and Ehmert, who were able to prove that when certain eruptions take place on the surface of the sun, the cosmic radiation incident on the earth suddenly gains in intensity. Thus, cosmic radiation is probably produced by big, periodically changing electromagnetic fields on the surfaces of the stars, and (according to Unsöld), on the surfaces of the many red stars in particular, or in intersteller space. These electromagnetic fields can assume very large magnitudes and accelerate charged particles to extremely high velocities (according to Bagge and Biermann, especially when a strong spot activity exists on the stellar surface). The hydrogen nuclei are the ones that are accelerated this way in the first place, for the stars consist mostly of hydrogen, but this effect is imparted also to the nuclei of heavier atoms; they leave the stars and speed through cosmic space as the *primary* cosmic radiation. The particles hit the atmosphere of the earth from outer space, with kinetic energies ranging mostly from 10^9 to 10^{10} electron-volts. But particles of energies up to 10^{16} electron-volts have also been observed. In the earth's atmospheric envelope, particles of such enormous energy produce not only atomic decay but also nuclear transmutations of all sorts, creating various kinds of elementary particles in so doing.

This is how the elementary particle just mentioned, the meson, was discovered (*Anderson*). The importance of this particle for nuclear physics is still uncertain. We know that its mass is approximately two hundred times the mass of an electron, that it carries one elementary charge of electricity, and that there are both positive and negative mesons. Beyond this, not much is known about its properties as yet. In 1947, Powell discovered another elementary particle, the mass of which is about three hundred times the mass of an electron. It is called *heavy meson* or π-particle. The researches of Leprince-Ringuet, Rochester and Butler have shown the probability of the existence of a still heavier elementary particle, of approximately nine hundred times the electronic mass.

Table Ic contains a list of all the above-mentioned elementary particles and their respective properties. By calling them 'elementary particles', we mean that they are not composed of still smaller particles, in contrast to the chemical atoms which obviously can be split up into component parts. But this does not indicate by any means that these elementary particles cannot be transformed. On the contrary, transformability is a characteristic of elementary particles. A photon can change into an electron plus a positron, and conversely, a photon can originate from an electron and a positron. But it